Lecture 2: Paradigms of Machine Learning

Supervised, Unsupervised & Sequential Learning Frameworks

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1. Executive Overview & Problem Taxonomy

Lecture 2 establishes a systematic classification framework for machine learning problems based on the **nature of feedback/supervision** available during training. Rather than viewing algorithms as isolated tricks, ML problems are categorized into three major learning paradigms: **Supervised**, **Unsupervised**, and **Sequential Learning**. Additionally, it maps real-world applications to these paradigms and defines the core mathematical prerequisites necessary for algorithmic modeling.

2. Primary Machine Learning Paradigms

A. Supervised Learning (Learning with a Teacher)

In supervised learning, the training dataset consists of input data points alongside explicit target **labels** (provided by a human expert or system). The objective is to learn a mapping function from inputs to labels that accurately predicts target outcomes on unseen test data.

- **Binary Classification:** Predicting one of two discrete classes (e.g., Spam vs. Non-Spam email filtering).
- **Multi-Class Classification:** Predicting one of finite K discrete classes (e.g., handwritten digit recognition 0–9).
- **Regression:** Predicting continuous real-valued outputs (e.g., forecasting rainfall in cm or housing prices).
- **Ordinal Classification / Ranking:** Predicting ordered categories or sorting items by preference (e.g., OTT movie recommendations).
- **Link Prediction:** Predicting missing connections in graph networks (e.g., LinkedIn friend/connection suggestions).

B. Unsupervised Learning (Discovering Latent Structure)

In unsupervised learning, data arrives without target labels or external guidance. The algorithm must uncover inherent patterns, natural clusters, or geometric structures directly from the data based on specified notions of similarity.

- **Clustering:** Grouping similar data points together without predefined labels (e.g., smartphones auto-grouping photos by face similarity).
- **Representation Learning:** Transforming raw input features (e.g., high-dimensional pixels) into lower-dimensional representations that simplify downstream learning.
- **Blind Source Separation:** Decomposing mixed overlapping signals into individual independent components (e.g., separating singer vocals from musical instruments in an audio file).

C. Sequential Learning (Iterative Decision-Making)

Unlike batch learning where all data is provided upfront, sequential learning involves an agent making decisions iteratively over time, receiving environment feedback, and updating its policy to optimize future outcomes.

- **Online Learning (Full Information):** Decision-maker receives feedback across all alternative options after each step (e.g., stock market prediction guided by expert advice).

- **Multi-Armed Bandits (Partial Feedback):** Feedback is only observed for the chosen action (e.g., clinical trial evaluation of a prescribed medication).
- **Reinforcement Learning:** An agent learns a *policy* mapping environmental states to actions via trial-and-error rewards (e.g., robot obstacle navigation).

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3. Real-World Application Taxonomy

The table below synthesizes real-world application examples discussed across the lectures, categorizing them by their overarching ML paradigm and specific problem type.

Application	ML Paradigm	Problem Sub-Category	Input / Output Structure
Email Spam Filter	Supervised	Binary Classification	Email content → {Spam, Non-Spam}
Rainfall Forecasting	Supervised	Regression	Weather metrics → Rainfall amount (cm)
OTT Movie Recommender	Supervised	Ordinal / Ranking	User profile → Ordered top-5 movie list
Social Connection Suggestion	Supervised	Link Prediction	Network graph → Probability of edge
Photo Album Face Grouping	Unsupervised	Clustering	Unlabeled photos → Face clusters
Audio Vocal/Instrument Separation	Unsupervised	Source Separation	Combined MP3 → Isolated vocal/audio tracks
Stock Market Trading	Sequential	Online Learning	Expert advice → Daily buy/sell decision
Clinical Treatment Decision	Sequential	Multi-Armed Bandit	Patient symptoms → Observed drug outcome
Autonomous Robot Navigation	Sequential	Reinforcement Learning	Environmental state → Action policy

4. Core Mathematical Prerequisites

Machine learning is built on three essential mathematical pillars that convert raw data into actionable decisions:

> **The Three Mathematical Pillars**
>
> **1. Linear Algebra (Structure):** Provides the language and geometric framework to represent data points, linear transformations, high-dimensional spaces, and projections.
>
> **2. Probability & Statistics (Uncertainty):** Supplies tools to quantify noise, incomplete measurement data, hidden variation, and real-world statistical confidence.
>
> **3. Optimization & Calculus (Decisions):** Defines loss/cost functions to measure model error and utilizes gradients to search for optimal model parameters.

5. Recommended Literature & Roadmap

- **Linear Algebra and Its Applications** – *Gilbert Strang* (Linear Algebra brush-up)
- **A First Course in Probability** – *Sheldon Ross* (Probability & Random Processes)
- **Pattern Recognition and Machine Learning (PRML)** – *Christopher M. Bishop* (Core reference textbook)
- **Mathematics for Machine Learning** – *Deisenroth, Faisal, & Ong* (Unified math foundations)